

Video Transcript

Video 5: Relational Data Structures (03:06)

So, here we're going to exercise both our Python writing skills and also our data structuring skills. So, this is typical data about a course that we're teaching, say, at MIT. So, we have the names of the students and we have the problem set number and the grade that the students get.

And then, of course, the Pset number needs to have another table that has, for example, the maximum score for that Pset. So, not every Pset has the same score. So, in this case, we'd have two tables, and the queries that we'd like to make from these tables are, for example, what's Fred's total for all the Psets. So, here we'd have Fred, and then we'd need to loop through and find Fred for Pset number 2 and Fred for Pset number 3, etc. So, we have to scan down this table to do that. Or what's Fred's average percentage for all the Psets? So, then we'd have to have Fred and scan down 1, 2, 3, etc, calculate the average, etc. So, there are questions that we'd like to ask of our data.

Now, an alternative to this would be, for example something like this, where we'd have the problem sets laid out in columns, so Fred and then Pset1, Pset2, Pset3, etc. And then, we'd still need this other related table with the PsetId and the MaxScore. Now, there's even another way we could lay this out, that we could have a table for students where the student has an ID. And in fact, that's what we do at MIT. We give every student an unique ID in case their name has a conflict.

And in my case, John Williams, there are three John Williams's in MIT that I know of, and there may be more. And so, we give an ID and typically we do this with all kinds of data. So, here we have the problem set, we have a problem PsetId with the grades. We might have a grade ID. Here we're not having a gradeld, but we're having a field. We're having 2 fields. A composite of these two will be unique, the studentId and the PsetId will be unique. So, in this case, we're going to have to look up the grade by the studentId and the PsetId. Let's take a look at what kind of code we might write in Python.